

CREATION OF ELECTRONICS ENCLOSURE

CSACE401

Create an Electronics Enclosure

Competence

RQF Level: 4

Learning Hours



70

Credits: 7

Sector: ICT and Multimedia

Trade: Computer System and Architecture

Module Type: Specific

Curriculum: ICTCSA4001 TVET Certificate IV in Computer System and Architecture

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Purpose statement	This specific module describes the skills, knowledge and attitude required to create an electronics enclosure. Upon completion of this module, the learner will be able to create a conceptual model of electronics enclosure, Design 3D Model of electronics enclosure with CAD, Print electronics enclosure.					
Learning assumed to be in place	▪ Apply SolidWorks Design ▪ Develop Embedded System hardware ▪ Setup LED and LCD screen					
Delivery modality	Training delivery		100 %	Assessment		Total 100%
	Theoretical content		20%	Formative assessment	20%	50%
	Practical work:		80%		80%	
	Group project and presentation	30%				
	Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competence and Performance Criteria

Element of competence	Performance criteria
1. Create a conceptual model of electronics enclosure	1.1. The electronic device enclosure requirements are properly identified based on the type of electronics being housed, and the specific design requirements
	1.2. Stages of 3D conceptual model are identified in response to the form factor of the electronic components.
	1.3. 3D conceptual design(sketch) is effectively developed according to circuit board diagrams specifications
2. Design 3D Model of electronics enclosure with CAD	2.1. Tools, material and equipment are appropriately identified based on ergonomics of electronic device shape and form factor.
	2.2. The CAD software is proficiently installed based on its manufacturer's installation guide.
	2.3. The 3D object is properly modelled based on conceptual design
3. Print electronics enclosure	3.1. The 3D printer is properly selected as per specific needs and requirements
	3.2. The 3D model is properly sliced as per slicing software(slicer) features and functionalities.
	3.3. The 3D printer is correctly set up based on their types and printing technologies
	3.4. The 3D model printing processes is properly applied according to the 3D model design.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Identify computer system components ✓ Describe of assemble and disassemble Computer system ✓ Identify computer system drivers and software ✓ Describe computer system transformation techniques	✓ Set a working place ✓ Select a computer system component ✓ Assemble and disassembling computer components ✓ Install a new hardware component ✓ Upgrade a computer hardware component ✓ Troubleshoot a system software ✓ Test a full functionality of computer system ✓ Elaborate a computer system report	✓ Innovative ✓ Organization ✓ Safety First ✓ Flexibility ✓ Time Management ✓ Problem Solving ✓ Forecasting ✓ Team work ✓ Hard worker ✓ Self-motivated

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Create a conceptual model of electronics enclosure 2. Design 3D model of electronics enclosure with CAD 3. Print electronics enclosure
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Learning outcome 1: Create a conceptual model of electronics enclosure	Learning hours: 15
Indicative content	
● Identification of enclosure modelling requirements ✓ Describe the electronics enclosure  Definition  The purpose ✓ Describe System requirements  Functional  Non-functional ✓ Interpret the electronic device diagrams ● Identification of object modelling stages ✓ Key factors to consider  Compatibility  Dimensional Fit  Thermal Management  Environmental protection  User Ergonomics  Aesthetics and Branding	

✚ Modularity and customization ✓ Design for EMI prevention/reduction ✚ EMI Shielding Design ✚ Grounding Schemes ✚ EMI filters ✚ Minimization of PCB layout ✓ Description of various modelling stages ✚ Skelton/Thumbnail ✚ Sculpting ✚ Texturing ✚ Rigging ✚ Animation • Sketch the conceptual design of the object ✓ Interpret electronic circuit board diagrams ✓ Apply Dimensioning of the object ✚ Selection of tools used in measuring dimensions ✚ Apply measurement ✓ Draw a conceptual model(ideation)	
Resources required for the learning outcome	
Equipment	▪ Computer ▪ Projector
Materials	▪ Internet ▪ Electricity ▪ Pen and Paper
Tools	▪ Sketching Software (Adobe Photoshop, Procreate, or Autodesk SketchBook) ▪ Mind Mapping Software (MindMeister, XMind, or Miro) ▪ Wireframing and Mockup Tools (Balsamiq, Adobe XD, or Figma).

	▪ 3D Modeling Software (Blender, Tinkercad, or Rhino) ▪ Concept Mapping Software (concept mapping tools like CmapTools) ▪ Photography and Image Editing Software (Adobe Photoshop) ▪ Presentation Software (Microsoft PowerPoint) ▪ Visual Diagramming Software (Lucidchart, draw.io, or Microsoft Visio) ▪ Measuring tools (calipers or digital Vernier calipers, Screwdrivers and festers, Heat gun or Hair Dryer, cutting tools (knives or scissors)) ▪ Assembly Tools (screws, nuts, bolts, ...)
Facilitation techniques or Learning activity	▪ Demonstration ▪ Individual and group work ▪ Practical exercise ▪ Group discussion
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Practical Assessments

Learning outcome 2: Design 3D Model of electronics enclosure with CAD	Learning hours: 35
Indicative content	
• Selection of tools, materials and equipment of 3D model ✓ Description of 3D model ✚ Definition ✚ Characteristics ✚ Types of angle views ✓ Required computer hardware specifications ✓ Software Tools ✚ SolidWorks ✚ AutoCAD ✚ Autodesk Maya • Installation of CAD software ✓ Selection criteria ✚ Computer hardware compatibility ✚ Software features ✓ Download CAD software ✓ installation manufacturer' s guide ✓ CAD environment installation process ✓ Rules of thumb ✓ Description of common elements in 3D software design interface ✚ Elements ✚ Interface Customizations ✓ 3D file types ✚ Stereolithography(.STL) ✚ Wavefront OBJ(.OBJ) ✚ 3D Manufacturing File(.3MF)	

✚ OLT Printing File(.OLTP) ✓ Description of 3D enclosure modeling considerations ✚ Shape and Size ✚ Wall Thickness and Material ✚ EMI Shielding Design ✚ Aesthetics and Branding ✚ Feedback and Iteration • Apply a 3D model design ✓ Interpret Conceptual model ✓ Design ✚ 3D Model ✚ Perform Rigging • Perform a 3D animation ✓ Keyframing ✚ Pose blocking ✚ Timing and spacing	
Resources required for the indicative content	
Equipment	▪ Computer
Materials	▪ Electricity ▪ Internet
Tools	▪ CAD software (SolidWorks, Autodesk Maya, AutoCAD)
Facilitation techniques or Learning activity	▪ Group discussion ▪ Demonstration ▪ Practical exercise
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Practical Assessments

Indicative content

- **Selection of 3D Printer**
 - ✓ Critical conditions of 3D printed enclosure technologies
 - ✚ Rapid prototyping (prototyping plastics)
 - ✚ High Temperature
 - ✚ High quality surface finish
 - ✚ High accuracy
 - ✚ Transparent
 - ✚ Fixed material
 - ✓ 3D Printing Technologies
 - ✚ FDM(FFF)
 - ✚ SLA and SLA flexible resin
 - ✚ SLS
 - ✚ Material jetting
 - ✚ Rubber-like plastics
- **Identification of printing materials**
 - ✓ EMI shielding
 - ✚ Conductive filament
 - ✚ Coatings
 - ✓ Thermoplastics
 - ✓ Printing properties
 - ✚ Print Quality
 - ✚ Print Speed
 - ✚ Print Volume
- **Slicing 3D model**
 - ✓ Definition
 - ✓ Identification of 3D high-quality 3D printing requirements

✓ Description of slicing software (Slicer) ✚ Types of slicing software ✚ Features ✚ Examples (Ultimaker Cura, Simplify3D, Slic3r / PrusaSlicer, MatterControl, CraftWare, ...) ✚ Selection criteria of slicer to be installed ✚ Install slicer ✓ Converting the 3D model into a series of 2D layers (slices) process ✚ Identify slicing software ✚ Import the 3d model ✚ Configure print settings ✚ Orientation and placement ✚ Support generation (if needed) ✚ Export g-code ✚ Transfer the g-code file to your 3d printer to print • Set up 3D printer for printing ✓ Identification of 3D printer settings ✓ Calibration of 3D printer settings using user manual • Perform 3D printing post-processing ✓ Finishing the enclosure ✓ Test quality assurance of the enclosure	
Resources required for the indicative content	
Equipment	▪ Computer ▪ 3D printer
Materials	▪ 3D printing Filaments (ABS, PLA, PETG, and polycarbonate), ▪ Adhesives ▪ Internet ▪ Electricity

	▪ Safety equipment (safety glasses, gloves) ▪ Post-processing equipment (sanding blocks, paint sprayers, or heat guns)
Tools	▪ CAD software (SolidWorks, Autodesk Maya, AutoCAD), Slicers software (Ultimaker Cura, PrusaSlicer, and Simplify3D)
Facilitation techniques or Learning activity	▪ Group discussion ▪ Demonstration ▪ Practical exercise
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Practical Assessments

Integrated/Summative assessment

Integrated situation

ABC Electronics is a small startup that specializes in developing innovative IoT devices for home automation. They have recently developed a new smart thermostat that allows homeowners to remotely regulate and control the temperature of a system. It is commonly used in heating, ventilation, and air conditioning (HVAC) systems to maintain a desired temperature level in a space Through a mobile app. However, they are facing a challenge in the final stages of product development of designing and implementing a suitable enclosure to house the electronic components of the thermostat. The enclosure needs to protect the sensitive electronics from dust, moisture, and physical damage while also being aesthetically pleasing and easy to manufacture. Therefore, The ABC Electronics wants to hire you as a solution provider to the above described problem of provided thermostat.

The task should be complete in 8 hours.

Resources

▪ Equipment		▪ Computer, 3D printer			
▪ Tools		▪ CAD software (SolidWorks, Autodesk Maya, AutoCAD), Slicers software (Ultimaker Cura, PrusaSlicer, and Simplify3D), calipers or digital Vernier calipers, Screwdrivers and festers, Heat gun or Hair Dryer, cutting tools (knives or scissors) ▪ Assembly Tools (screws, nuts, bolts, ...)			
▪ Materials		▪ 3D printing Filaments (ABS, PLA, PETG, and polycarbonate), adhesives, ▪ Internet, electricity ▪ Paper and pens ▪ Safety Equipment (Safety glasses, gloves), ▪ Post-Processing Equipment (sanding blocks, paint sprayers, or heat guns),			
Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
1: Create a conceptual model of an electronic enclosure (15%)	1. Identification of the electronic device enclosure requirements	Electronic circuit diagram is interpreted			5
		Appropriate measurement tools are selected			2
		Accurate Dimensions are measured			5
	2. drawing 3D conceptual	Appropriate drawing tools are selected			2

	design(sketch)	Components heat is managed			1
Learning outcome 2: Design a 3D Model of an electronics enclosure with CAD (50%)	1. Identification of tools, materials, and equipment	Conceptual model is interpreted			10
		Appropriate CAD software is selected			3
		Appropriate hardware is selected			2
	2. Installation of CAD software	CAD software is installed			5
	3. Designing of 3D object	3D model is designed			20
		3D animation is performed			10
Learning outcome 3: Print electronics enclosure (35%)	1. Selection of 3D printer	3D Printer is selected			2
		Printing materials are selected			2
	2. Slicing 3D model	Slicer software is installed			5
		3D model is sliced			5
	3. Configure 3D printer	The 3D printer is configured			10
	4. Finishing Electronics Enclosure (post - processing)	Surface Finishing is performed			2
		Gasketing is performed			2
		Adhesive is applied			2

		EMI/RFI Shielding is applied			3
		Sealing and Coating is applied			2
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

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